CYBERSECURITY LOG ANALYSIS REPORT

Student Answer Template

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Activity: CTM-ACTIVITY-8

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Instructions: Analyze each CSV dataset provided in the GitHub repository. Identify the likely attack type, cite specific log evidence, assess whether the attack succeeded, and explain the possible impact if the attack were successful. Do not rely on filenames or guesses; support your conclusions with the log entries.

ATTACK 1 ANALYSIS

1. Identification of Attack

2. Evidence from the System Logs

1. Identification of Attack

- **Attack Type:** SSH Brute-Force / Credential Stuffing Attack.
- **Characteristics & Justification:** The log entries show repeated authentication failures over TCP port 22 (SSH) targeting host 192.0.2.10 from a single external IP address (198.51.100.23) within seconds of each other. The attempts cycle through standard high-privilege usernames (admin, administrator, root).

2. Evidence from the System Logs

Evidence 1

- **Event ID / Timestamp:** BF-0001 to BF-0004 | 2026-10-31T08:14:00 to 2026-10-31T08:14:10.
- **Observed Activity:** Sequential failed login attempts targeting usernames admin, administrator, and root from source IP 198.51.100.23.
- **Why It Matters:** Confirms an automated script testing common credentials against an open SSH management port.

Evidence 2

- **Event ID / Timestamp:** BF-0005 | 2026-10-31T08:14:20.
- **Observed Activity:** Logged event account_lockout with status 5 blocked and severity critical.
- **Why It Matters:** Shows that host defensive threshold controls actively engaged to temporarily block the targeted user account after reaching five consecutive password failures.

Evidence 3

- **Event ID / Timestamp:** BF-0007 | 2026-10-31T08:16:30.
- **Observed Activity:** Logged event rate_limit_triggered with status 6 blocked and details indicating the source IP was rate-limited.
- **Why It Matters:** Proves perimeter defenses actively restricted further inbound connections from the attacker's IP address.

3. Attack Analysis

An attacker operating from IP `198.51.100.23` executed a high-frequency credential guessing script against SSH server `192.0.2.10` on port 22. The script cycled through default administrative accounts. System security rules successfully intercepted the brute-force attempt by triggering an account lockout at attempt 5 and

subsequently enforcing network rate-limiting against the attacking IP address. Meanwhile, normal authentication from legitimate user `student` (IP `203.0.113.44`, Event `BF-0008`) was allowed, proving system operational health.

4. Attack Status

Status: Unsuccessful.

Justification: All entries associated with the attacker's IP address (198.51.100.23) resulted in failed_login, account_lockout, or rate_limit_triggered statuses. No successful login event occurred from the attacker's source address.

5. Possible Scenario if the Attack Succeeds

Target System Impact: Full remote command-line access with elevated administrative/root privileges.

Compromised Assets: Local host data, configuration files, system credentials, SSH keys, and system services.

Consequences for Users & Organization: Attacker gains persistent system access, ability to install malicious software or ransomware, potential lateral movement to internal network segments, and data exfiltration.

6. Prevention and Mitigation

Recommendation 1: Disable Root Password Authentication & Reposition SSH: Disable remote password login for `root` in `/etc/ssh/sshd_config` (`PermitRootLogin no`) and bind SSH to a non-standard network port.

Recommendation 2: Enforce Public-Key Authentication & MFA: Require encrypted SSH key pairs with passphrase protection alongside Multi-Factor Authentication (MFA) for remote management access.

Recommendation 3: Deploy Dynamic IP Blocking (e.g., Fail2ban): Configure automated intrusion prevention rules to drop IP packets immediately after 3 consecutive failed login attempts before lockout thresholds are reached.

ATTACK 2 ANALYSIS

1. Identification of Attack

Attack Type: Spear Phishing & Malicious Payload Delivery Campaign.

Characteristics & Justification: Involves external inbound emails attempting to impersonate internal departments (it-support@example.test, payroll@example.test) while containing deceptive web domains (account-verify.example.test, payroll-portal.example.test) and executable HTML attachments (payroll_update.html) targeted at specific organization staff.

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: `PH-0002` & `PH-0003` | `2026-10-31T09:03:40` to `2026-10-31T09:05:10`.
Observed activity: `PH-0002` logged `suspicious_url_detected` for domain `account-verify.example.test`, followed by `PH-0003` recording `link_clicked` by `employee01@example.test`.
Why it matters: Direct proof that an employee interacted with an unverified external hyperlink sent via email.

Evidence 2: Event ID / Timestamp: `PH-0004` | `2026-10-31T09:05:40`
Observed activity: Event `credential_form_detected` for domain `account-verify.example.test` with

action set to `blocked`.
Why it matters: Shows that the secure web gateway proactively blocked the phishing login portal before credential submission could take place.

Evidence 3: Event ID / Timestamp: `PH-0007` | `2026-10-31T09:09:00`
Observed activity: Event `attachment_scanned` for recipient `employee02@example.test` with payload `payroll_update.html` resulting in status `blocked`
Why it matters: Confirms automated mail filters caught and stopped a suspicious HTML payload from reaching the end user.

3. Attack Analysis

An external entity launched a targeted phishing campaign spoofing internal support and payroll identities. Although `employee01` clicked the provided link (`PH-0003`), the underlying credential harvesting page was blocked by web security controls (`PH-0004`). Subsequent email vectors targeting `employee01` and `employee02` were automatically quarantined or had their attached HTML files blocked (`PH-0005`, `PH-0007`).

4. Attack Status

Status: Unsuccessful.

Justification: Although a user clicked the phishing link (PH-0003), security systems blocked access to the credential submission form (PH-0004), quarantined the message (PH-0005), and prevented execution of the malicious attachment (PH-0007). No credentials or system access were compromised.

5. Possible Scenario if the Attack Succeeds

Target System Impact: Compromise of corporate user credentials and unauthorized access to identity providers (SSO/SaaS portals).

Compromised Assets: User email access, corporate communications, internal directories, payroll records, and identity tokens.

Consequences for Users & Organization: Business Email Compromise (BEC), unauthorized direct-deposit payroll modifications, internal phishing originating from legitimate accounts, and loss of confidential data.

6. Prevention and Mitigation

Recommendation 1: Implement Email Authentication Controls (SPF, DKIM, DMARC): Enforce strict `DMARC` policies (`p=reject`) to prevent external servers from spoofing internal domains like `it-support@example.test`

Recommendation 2: Deploy Secure Email & Web Gateways with Link Sandboxing: Route inbound email traffic through inspection tools that analyze, rewrite, and sandbox links and HTML/script attachments in real time.

Recommendation 3: Conduct Ongoing Security Awareness & Phishing Simulations: Regularly train staff to identify external domain mismatches, suspicious file attachments (`.html`, `.hta`), and urgent requests asking for credential verification.

ATTACK 3 ANALYSIS

1. Identification of Attack

Attack Type: Distributed Denial of Service (DDoS) / Application-Layer HTTP Flood Attack.

Characteristics & Justification: Rapid request volume spikes on port 443 (HTTPS) directed at server 192.0.2.20 originating across multiple distinct IP addresses (198.51.100.90, 198.51.100.91, 198.51.100.92, 198.51.100.93). The request volume far exceeded standard baselines (35 req/sec vs 120–850 req/sec), triggering HTTP 503 service outage errors.

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: `DOS-0002` & `DOS-0003` | `2026-10-31T10:20:10` to `2026-10-31T10:20:20`
Observed activity: Request rates surged to 420 and 610 req/sec against a normal baseline of 35 req/sec, causing the web server to return HTTP response code `429` (`Too Many Requests`)
Why it matters: Identifies an abnormal volumetric flooding attempt overloading application processing capacity.

Evidence 2: Event ID / Timestamp: `DOS-0004` & `DOS-0005` | `2026-10-31T10:20:30` to `2026-10-31T10:20:40`
Observed activity: Events `service_latency` and `availability_check` logged HTTP response code `503` (`Service Unavailable`) with details stating `Health check failed` as requests reached 850 req/sec.
Why it matters: Direct evidence of complete service disruption and server unresponsiveness during the peak traffic window.

Evidence 3: Event ID / Timestamp: `DOS-0006` & `DOS-0008` | `2026-10-31T10:20:50` to `2026-10-31T10:22:00`
Observed activity: Event `mitigation_enabled` activated traffic filtering, stabilizing system health checks back to HTTP code `200` with status `resolved`
Why it matters: Documents the activation of defensive filtering controls that successfully mitigated the volumetric attack and restored application services.

3. Attack Analysis

A distributed set of IP addresses launched an HTTPS application flood targeting web host `192.0.2.20`. Request volume quickly escalated from 120 req/sec to 850 req/sec against a normal baseline of 35 req/sec. Initial rate-limiting (HTTP 429) was insufficient, leading to temporary resource exhaustion and HTTP 503 errors (`DOS-0004`, `DOS-0005`). Traffic filtering controls engaged at 10:20:50 (`DOS-0006`), dropping malicious traffic and returning server availability to normal levels by 10:22:00 (`DOS-0008`)

4. Attack Status

Status: Partially Successful.

Justification: The attack successfully caused temporary service downtime and unavailability, evidenced by failed health checks and HTTP 503 server errors (DOS-0004, DOS-0005), prior to automated mitigation rules restoring full service stability (DOS-0008).

5. Possible Scenario if the Attack Succeeds

Target System Impact: Prolonged web server unresponsiveness, connection timeouts, and system resource (CPU/RAM/bandwidth) exhaustion.

Compromised Assets: System availability, bandwidth limits, baseline SLA commitments, and operational uptime (no data integrity or confidentiality loss).

Consequences for Users & Organization: Inability for clients to reach corporate web services, loss of business revenue, customer frustration, SLA penalties, and incident response overhead.

6. Prevention and Mitigation

Recommendation 1: Deploy Cloud-Based DDoS Scrubbing Services: Utilize Anycast DDoS mitigation networks (e.g., Cloudflare, AWS Shield) to absorb high-volumetric HTTP floods before they reach origin infrastructure.

Recommendation 2: Configure Adaptive Rate Limiting & Web Application Firewall (WAF): Set adaptive request thresholds on reverse proxies and load b**alancers to automatically throttle or challenge anomalous request bursts.**

Recommendation 3: Implement Autoscaling Infrastructure & Anycast Load Balancing: Establish scalable compute groups coupled with load balancers to distribute traffic loads across multiple instances while filtering rules drop bad traffic vectors.

FINAL REFLECTION

1. Among the three attacks, which could cause the greatest damage to an organization? Explain your answer.

-**Phishing (Attack 2)** has the potential to inflict the most severe long-term damage on an organization. SSH brute-force attacks target exposed system ports, and DDoS attacks disrupt availability temporarily. In contrast, phishing targets human users to bypass perimeter defenses entirely.

2. Why is analyzing system logs important in detecting and responding to cybersecurity incidents?

Early Incident Detection: System logs provide continuous monitoring across firewalls, servers, and applications, allowing security teams to spot anomalous behavior (e.g., high failure rates, unusual request spikes) before major damage occurs.

Forensic Reconstruction: Logs contain precise timestamped evidence (source IPs, event IDs, response codes) necessary to determine *how* an attack occurred, *what* endpoints were targeted, and *how far* the attacker progressed.

Verification of Attack Outcome: Log analysis helps analysts differentiate between benign traffic, automated attack noise, blocked attempts, and actual compromises.

3. What is the most important cybersecurity lesson you learned from this activity?

The key takeaway from this activity is the principle of **Defense-in-Depth and Verification over Assumption**.

An attack attempt (such as a clicked link or a brute-force failure) does not automatically mean a system has been compromised. Security monitoring requires looking beyond initial alert triggers to evaluate status codes and defensive responses. Implementing multi-layered defenses—such as rate-limiting, web gateway filters, and account lockouts—ensures that if one layer fails, secondary controls can prevent a full breach.

SUBMISSION CHECKLIST

☐ I analyzed all three CSV datasets.

☐ I identified each attack using evidence from the logs.

☐ I included at least three supporting log observations for each attack.

☐ I justified the status of each attack.

☐ I explained a possible successful-attack scenario for each attack.

☐ I provided at least three relevant security recommendations for each attack.

☐ I completed the final reflection.

Submission filename: Surname_Firstname_Cybersecurity_Log_Analysis.docx

Reminder: Distinguish between an attack attempt, detection, prevention, and actual compromise. A suspicious event or a clicked link does not automatically prove that an account or system was compromised. Base your conclusion on the available evidence.